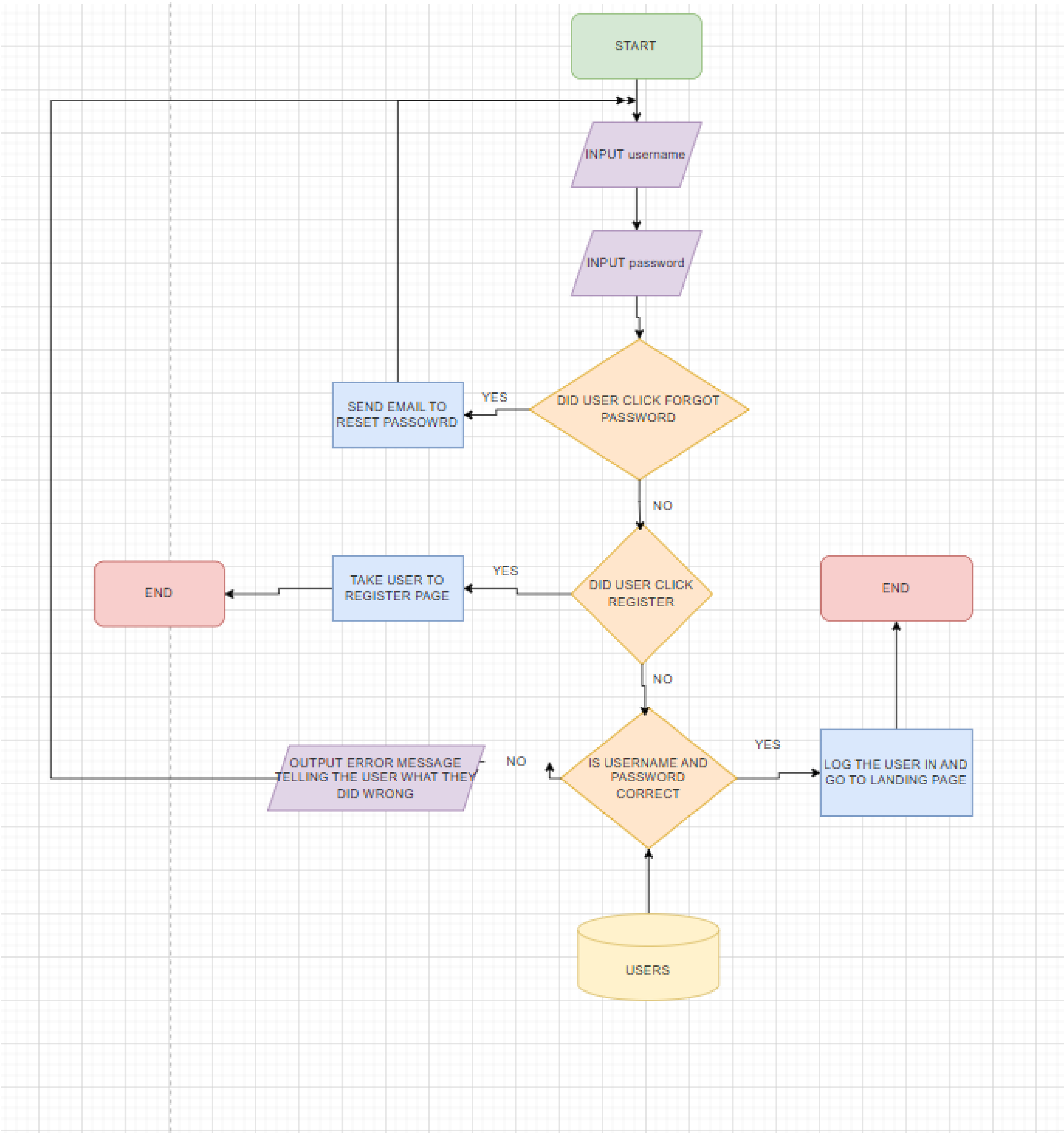
Contents

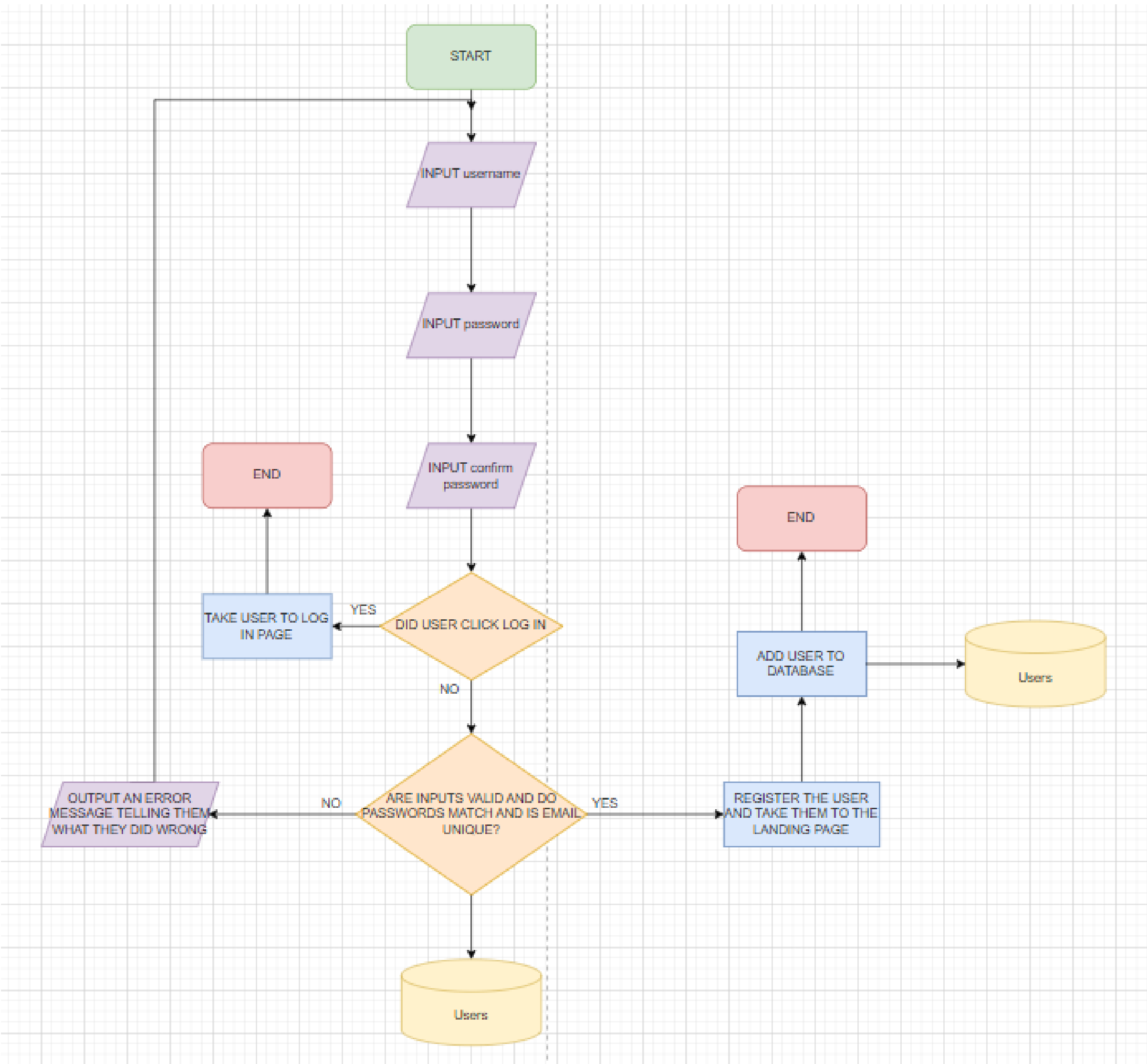
FLOWCHARTS

Log in



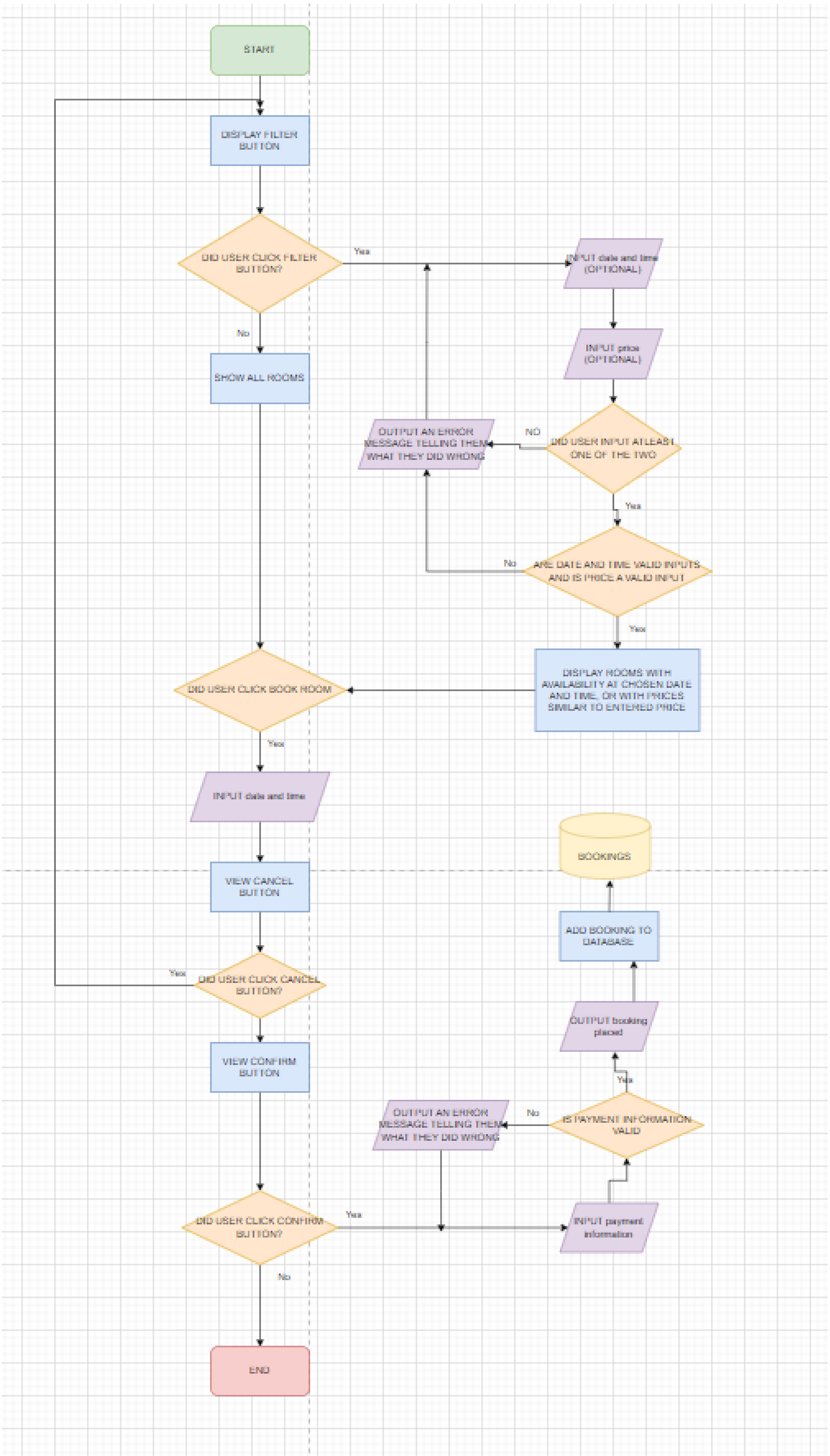
The user will be prompted to enter their username and password the system will then check if the user clicked other options like forgot password or register if they did it will take the appropriate actions like sending a reset email link or taking the user to the register page if not the system will check if the username and password match the ones in the database and if not, not, it will send an error message telling the user what they did wrong if its correct the system will log the user in return them to the landing page.

Register



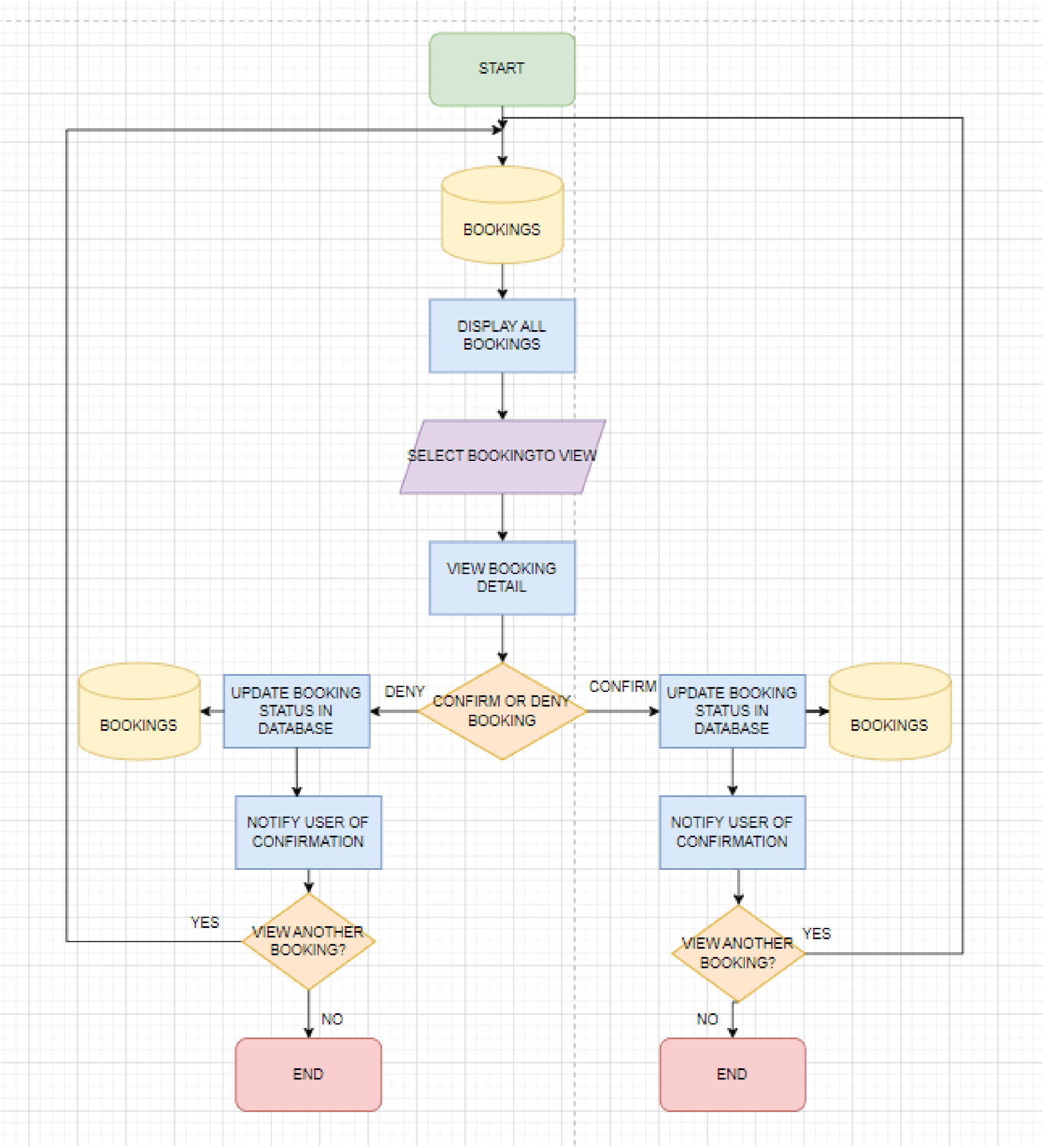
The user will be prompted to enter their username and password and password again to confirm it, the system then checks if the two passwords match and if the email is unique to any other account from the database, then they can proceed to add the user to the database if everything is good but if there are any issues the system prompts the user with an error message explaining what they have done wrong and asks them to try gaain.

Making a booking



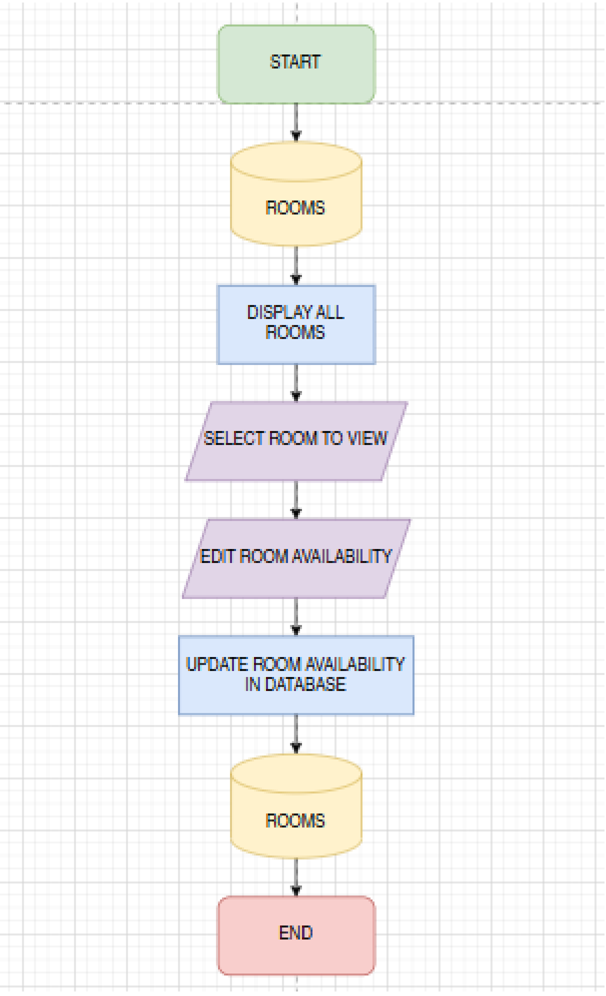
The page loads up showing all rooms available to book and also a filter button, tif the user chooses to filter the rooms they are asked to filter by price or by availability at a certain date and time, the user must choose one of the two and the system displays the necessary rooms in which they have filtered to. When the user finds the room that they want they can click the book room button, enter the details of their booking and click confirm, there they will be asked to pay and if their payment is successful, they will receive a booking confirmation message and the booking will be added to the database. The system is quipped with fail safes meaning if the user enters the wrong things at any time the user is given an error message explaining what went wrong and are prompted to try again.

Staff confirming or denying bookings

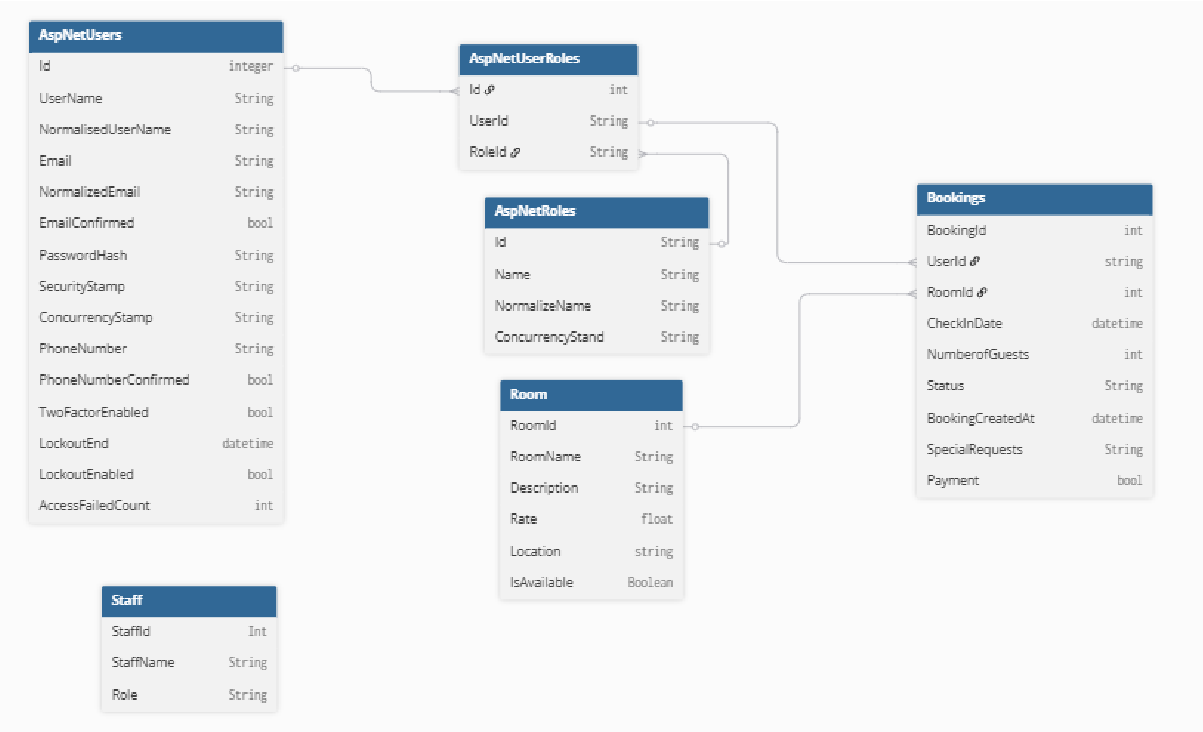


Staff are allowed to confirm and deny bookings, so this function makes that possible by accessing the database to be able to display the full list of all the bookings onto the web page, the staff member then finds a booking they want to see and selects it. The eb page displays the details of the booking that they have selected and they are presented with two buttons, a confirm or deny button, clicking either will update the booking information on the database, notify the user that their booking has either been accepted or denied and then the system asks the staff member if they would like the move on to another booking, clicking yes will take them back to the page with the full list of all bookings made.

Editing room availability



Database Design



Data Dictionary

AspNetUsers

Column	***Data Type***	***Required***	***Default***	***Validation***	***Key Type***
Id	String	Yes			Primary
UserName	String	Yes			None
NormalizedUserName	String	Yes			None
NormalizedEmail	String	Yes		Must include a valid email structure, must have a”@” symbol, must have a domain name	None
EmailConfirmed	String	Yes			None
PasswordHash	Bool	Yes		Users must confirm their email	None
SecurityStamp	String	Yes			None
ConcurrencyStamp	String				None
PhoneNumber	String				None
PhoneNumberConfirmed twoFactorEnabled	String				None
LockoutEnd	Bool				None
LockoutEnabled	DateTime				None

AccessFailedCount	Int				None

AspUserRoles

Column	***Data Type***	***Required***	***Default***	***Validation***	***Key Type***
Id	Int	Yes			Primary
UserId	String	Yes	Users Id		Foreign
RolesId	String	Yes			None

Bookings

Column	***Data Type***	***Required***	***Default***	***Validation***	***Key Type***
BookingId	Int	Yes	BookingId	Must be a numerical value	Primary
UserId	String	Yes	UserId		Foreign
RoomId	Int	Yes			Foreign
CheckInDates	DateTime	Yes		Must be a valid date and time	None
NumberOfGuests	Int	Yes		Must be a numerical value, CAN NOT BE LESS THAN 0	None
Status	String	Yes			None
BookingCreatedAt	DateTime	Yes		Must be a valid date and time	None
SpecialRequests	String	No		Can not be more than 500 characters	None
Payment	Boolean	Yes			None

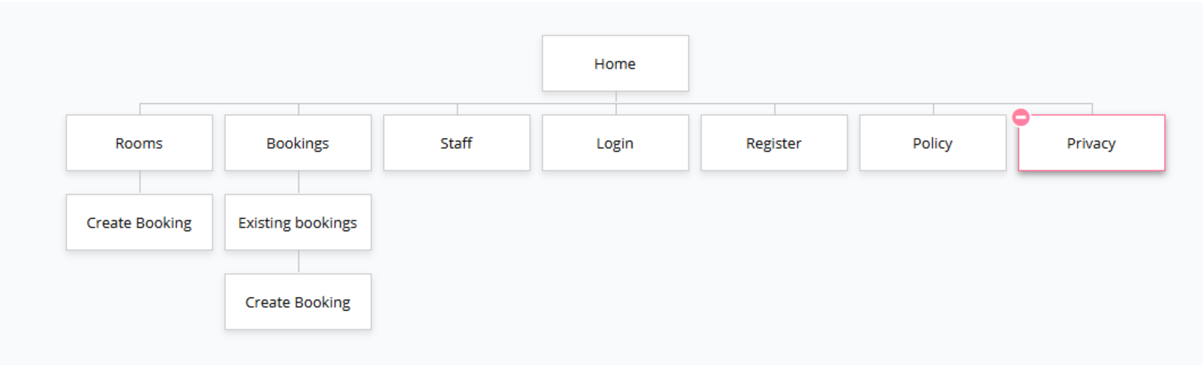
Rooms

Column	***Data Type***	***Required***	***Default***	***Validation***	***Key Type***
RoomId	Integer	Yes	RoomId		Primary
RoomName	String	Yes			None
Description	String	Yes			None
Rate	Float	Yes		Must be a numerical value, Must be a float value, Must have a currency	None
Location	String	Yes			None
IsAvailable	Bool	Yes			None

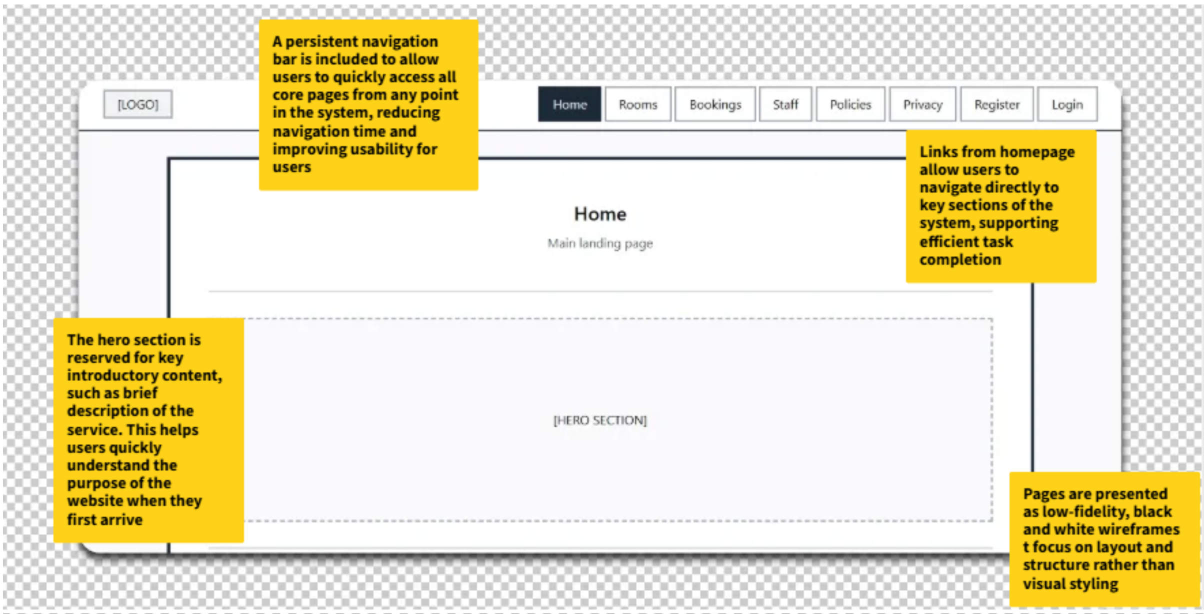
Staff

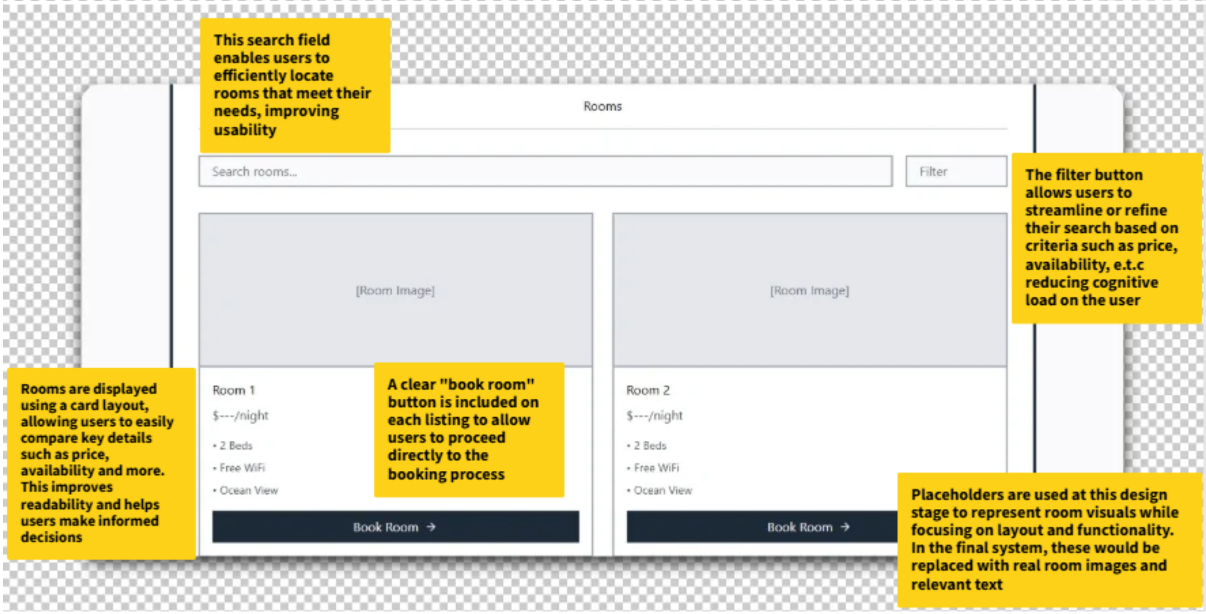
Column	*Data Type*	*Required*	*Default*	*Validation*	*Key Type*
StaffId	int	Yes	StaffId		Primary
Name	String	Yes			None
Role	String	Yes			None

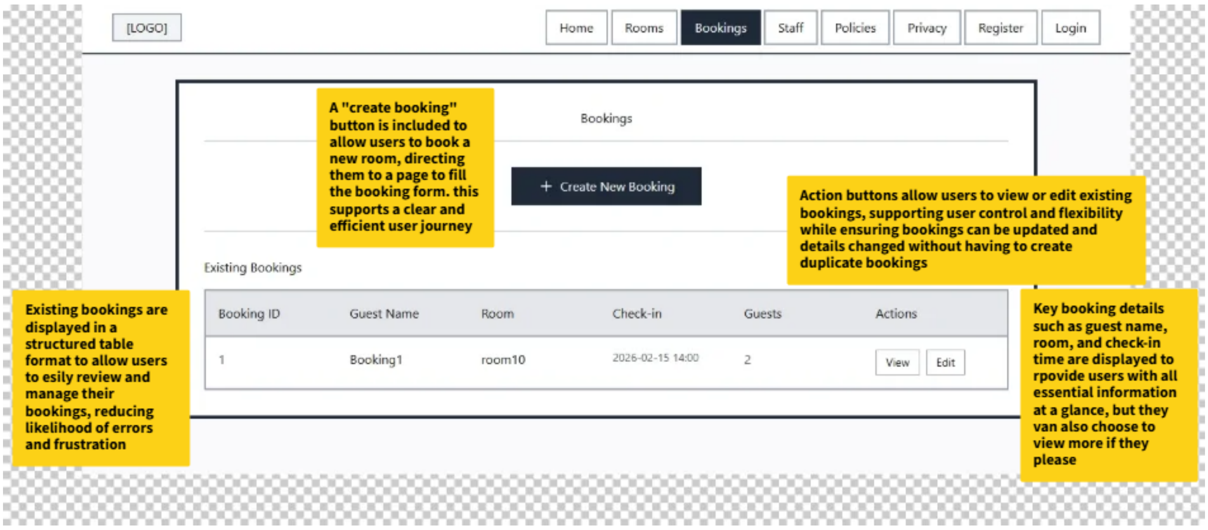
Sitemap

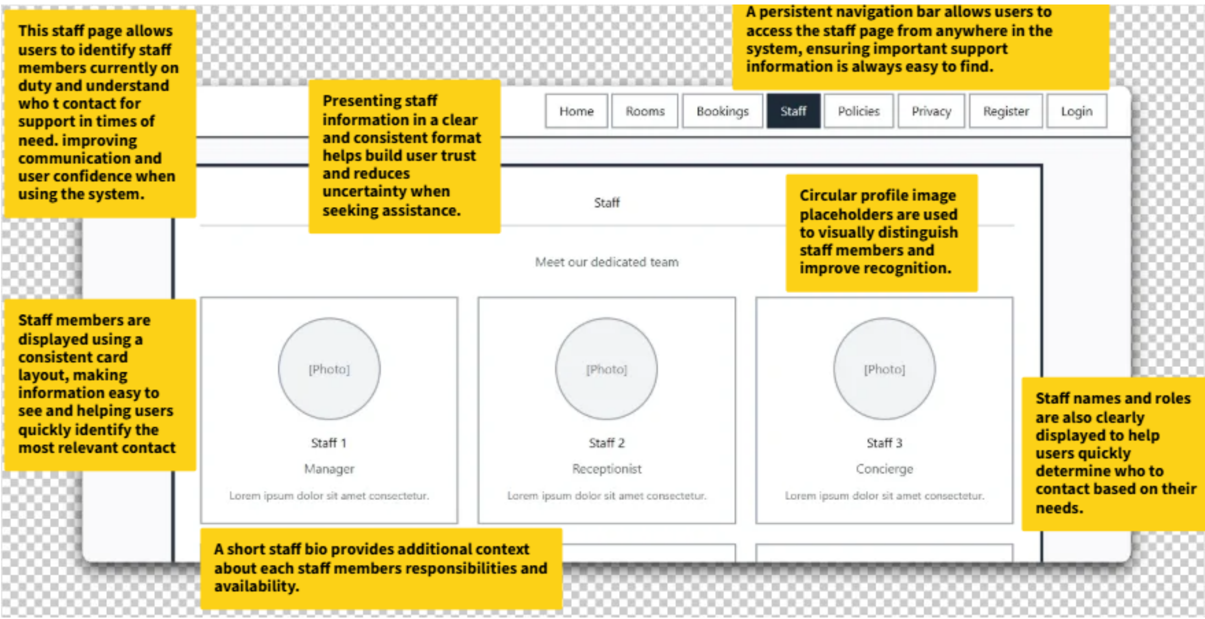


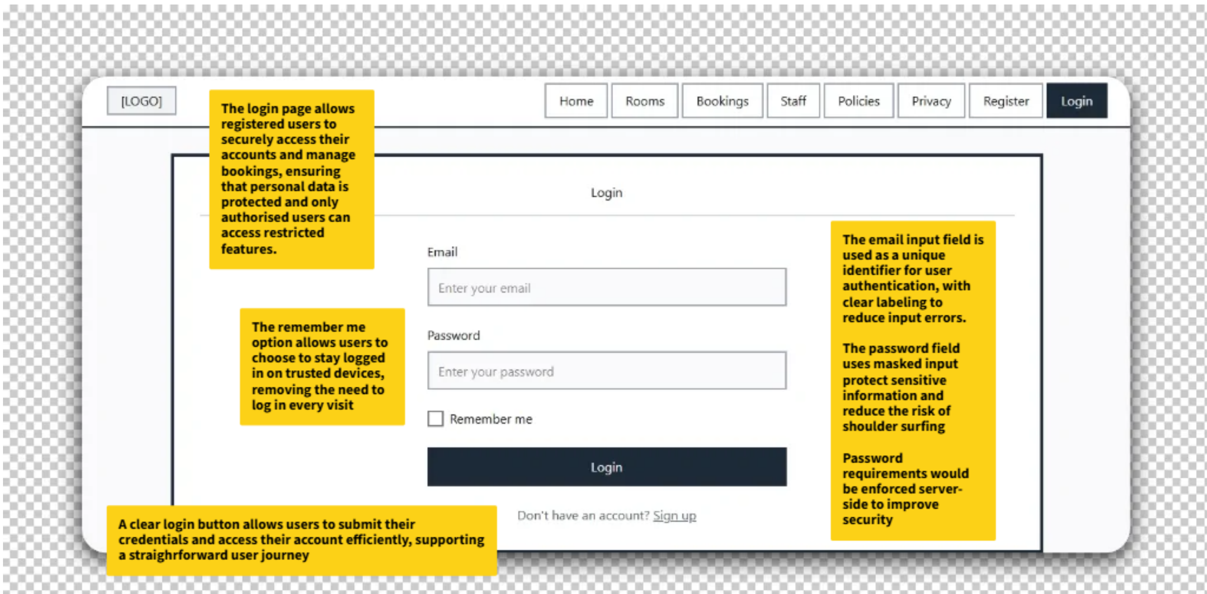
WireFrames

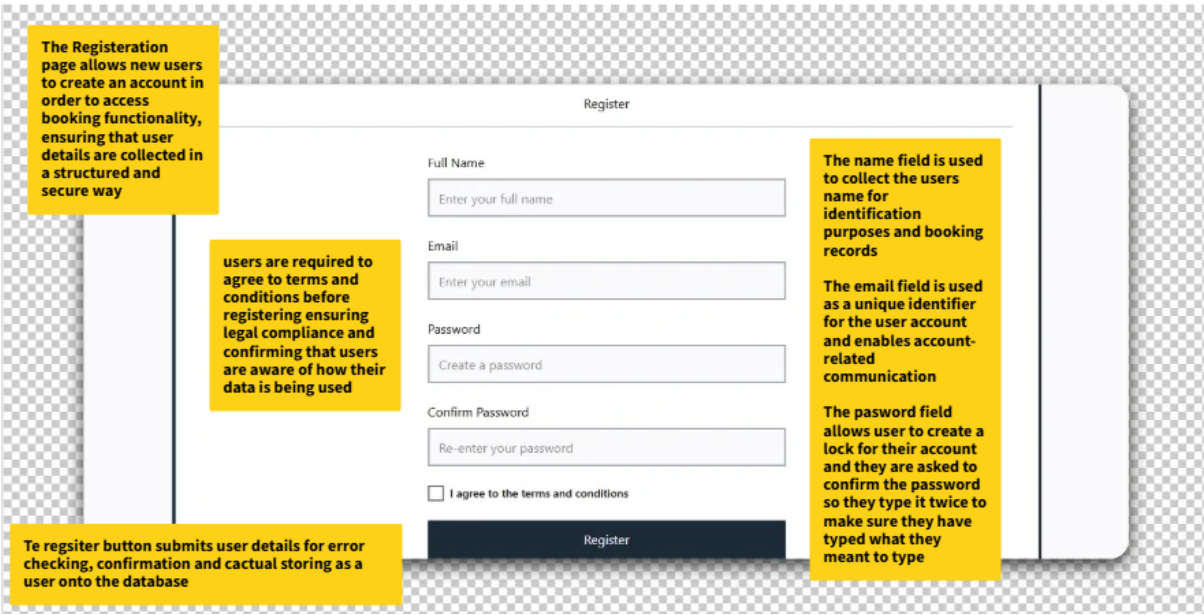

[LOGO]
Home
Rooms
Bookings
Staff
Policies
Privacy
Register
Login
A persistent navigation bar is included to allow users to quickly access all core pages from any point in the system, reducing navigation time and improving usability for users
Links from homepage allow users to navigate directly to key sections of the system, supporting efficient task completion
Home
Main landing page
The hero section is reserved for key introductory content, such as brief description of the service. This helps users quickly understand the purpose of the website when they first arrive
[HERO SECTION]
Pages are presented as low-fidelity, black and white wireframes t focus on layout and structure rather than visual styling

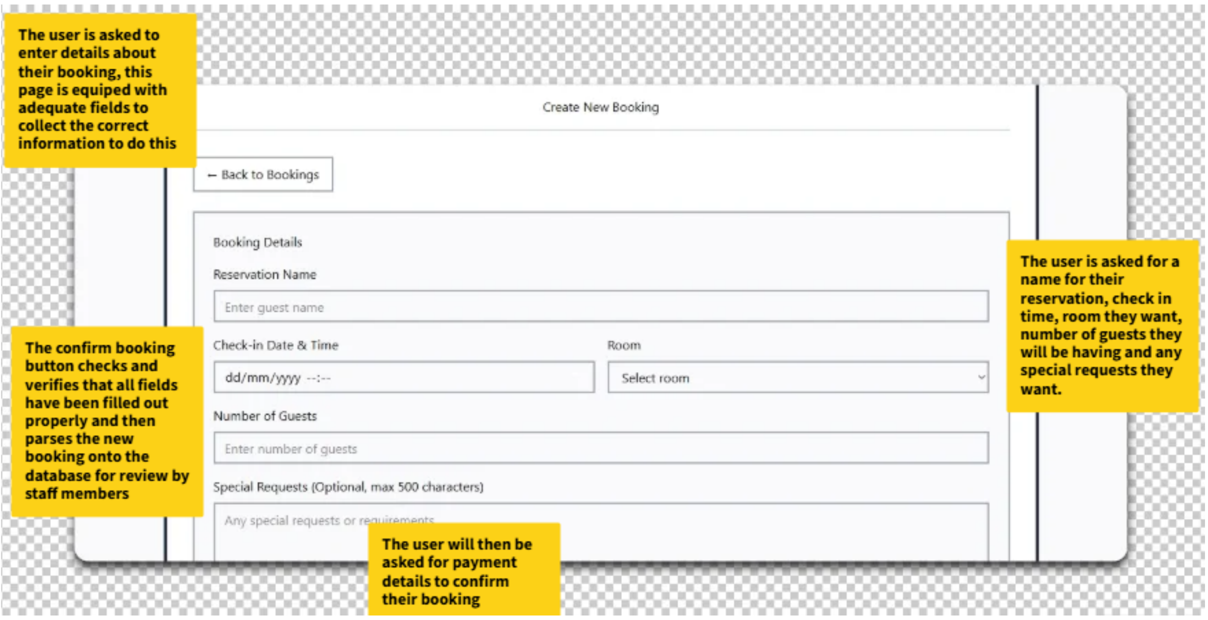

This search field enables users to efficiently locate rooms that meet their needs, improving usability
Rooms
Search rooms...
Filter
The filter button allows users to streamline or refine their search based on criteria such as price, availability, e.t.c reducing cognitive load on the user
[Room Image]
[Room Image]
Rooms are displayed using a card layout, allowing users to easily compare key details such as price, availability and more. This improves readability and helps users make informed decisions
Room 1
$---/night
• 2 Beds
• Free WiFi
• Ocean View
A clear "book room" button is included on each listing to allow users to proceed directly to the booking process
Room 2
$---/night
• 2 Beds
• Free WiFi
• Ocean View
Book Room
Book Room
Placeholders are used at this design stage to represent room visuals while focusing on layout and functionality. In the final system, these would be replaced with real room images and relevant text


[LOGO]
Home
Rooms
Bookings
Staff
Policies
Privacy
Register
Login
A "create booking" button is included to allow users to book a new room, directing them to a page to fill the booking form. this supports a clear and efficient user journey
Bookings
Create New Booking
Action buttons allow users to view or edit existing bookings, supporting user control and flexibility while ensuring bookings can be updated and details changed without having to create duplicate bookings
Existing Bookings
Booking ID
Guest Name
Room
Check-in
Guests
Actions
1
Booking1
room10
2026-02-15 14:00
2
View
Edit
Existing bookings are displayed in a structured table format to allow users to esily review and manage their bookings, reducing likelihood of errors and frustration
Key booking details such as guest name, room, and check-in time are displayed to rpovide users with all essential information at a glance, but they van also choose to view more if they please

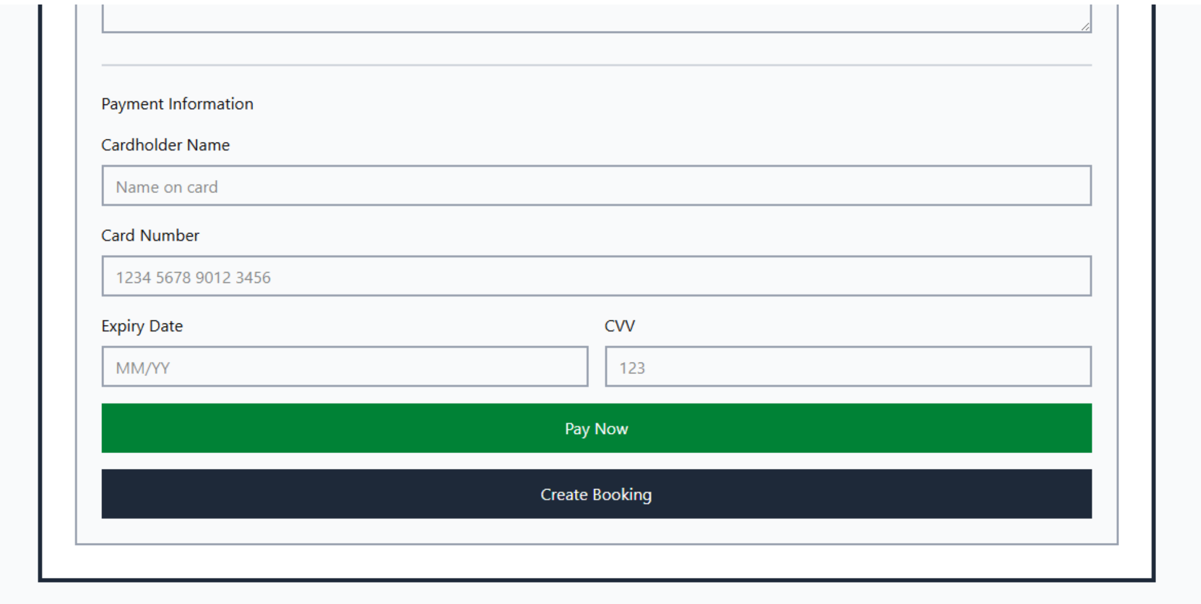

This staff page allows users to identify staff members currently on duty and understand who t contact for support in times of need. improving communication and user confidence when using the system.
Presenting staff information in a clear and consistent format helps build user trust and reduces uncertainty when seeking assistance.
A persistent navigation bar allows users to access the staff page from anywhere in the system, ensuring important support information is always easy to find.
Home
Rooms
Bookings
Staff
Policies
Privacy
Register
Login
Staff
Circular profile image placeholders are used to visually distinguish staff members and improve recognition.
Meet our dedicated team
Staff members are displayed using a consistent card layout, making information easy to see and helping users quickly identify the most relevant contact
[Photo]
Staff 1
Manager
Lorem ipsum dolor sit amet consectetur.
[Photo]
Staff 2
Receptionist
Lorem ipsum dolor sit amet consectetur.
[Photo]
Staff 3
Concierge
Lorem ipsum dolor sit amet consectetur.
Staff names and roles are also clearly displayed to help users quickly determine who to contact based on their needs.
A short staff bio provides additional context about each staff members responsibilities and availability.

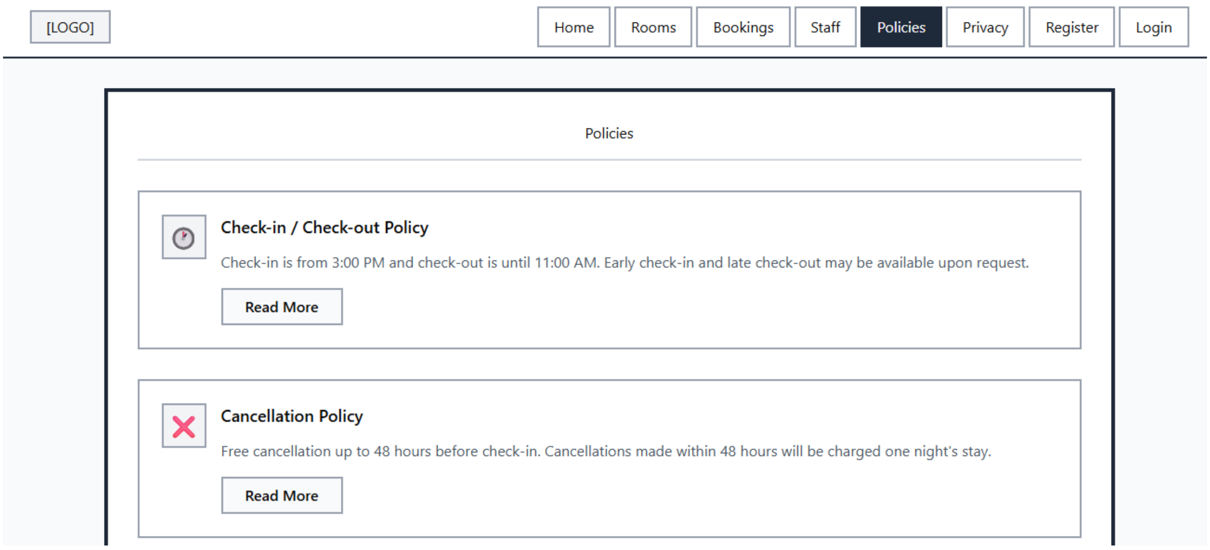

[LOGO]
The login page allows registered users to securely access their accounts and manage bookings, ensuring that personal data is protected and only authorised users can access restricted features.
Home
Rooms
Bookings
Staff
Policies
Privacy
Register
Login
Login
Email
Enter your email
The email input field is used as a unique identifier for user authentication, with clear labeling to reduce input errors.
The remember me option allows users to choose to stay logged in on trusted devices, removing the need to log in every visit
Password
Enter your password
The password field uses masked input protect sensitive information and reduce the risk of shoulder surfing
Remember me
Login
Password requirements would be enforced server-side to improve security
A clear login button allows users to submit their credentials and access their account efficiently, supporting a straighrforward user journey
Don't have an account? Sign up


The Registeration page allows new users to create an account in order to access booking functionality, ensuring that user details are collected in a structured and secure way
Register
Full Name
Enter your full name
The name field is used to collect the users name for identification purposes and booking records
users are required to agree to terms and conditions before registering ensuring legal compliance and confirming that users are aware of how their data is being used
Email
Enter your email
The email field is used as a unique identifier for the user account and enables account-related communication
Password
Create a password
Confirm Password
Re-enter your password
The pasword field allows user to create a lock for their account and they are asked to confirm the password so they type it twice to make sure they have typed what they meant to type
I agree to the terms and conditions
Register
Te regsiter button submits user details for error checking, confirmation and cactual storing as a user onto the database



(part of booking page above)





The policies page allows users to be able to read in on the poliies of the company so they know what to expect, these policies could include things like refund policies, checkin and check out policies and many more

This page features different policies in a card format, has a general image to explain the policy, a body of text stating the policy and a read more button if they choose to find out more about that specific policy. This page helps build user to company trust as they now have better communications about “rules and regulations” and how things work and are carried out in the company.

Test Strategy

Date of Test	***Component to be tested***	***Type of test to be carried out***	***Prerequisites and dependencies***
30/01/2026	User Register system	White box testing, black box testing, functionality testing, Security testing, Integration Testing.	The register function should be fully functional and implemented. Migrations and update database must be applied. The controller logic should be completed to handle user registration and error handling for incorrect inputs duplicate accounts and password complexity must be set up. Tests should include valid/invalid Test data needed. The log-in page must be created, and the migration and update

			database must be completed. There must be working controller logic. Testers must test valid/invalid credentials, session management, variability's (SQL injection) and password encryption. Example of test data: - Normal: Email: “Test@teste mail.com,” Password: "Password123 ?”, Confirm Password: “Password123 ?” - Erroneous: Email: “falseemail” (Should contain a “@ symbol”, Should contain a domain e.g. “.com”), Password: “111” (Should be longer, should be a string, should have a capital letter and a symbol), Confirm Password: “1er” (Dose not match original password) - Extreme: test@testema il.com, Password: “QwE12?” (Exactly 6 characters which is the minimum and includes 1 symbol which is the minimum)
2/02/2026	User log in system	White box testing, black box testing, functionality testing, Security testing, Integration Testing	Test data needed. The log-in page must be created, and the migration and update database must be completed. There must be working controller logic. Testers must test valid/invalid credentials, session management,

			variability's (SQL injection) and password encryption. Example of test data: - Normal: Email: “Test@teste mail.com”, Password: “Password123 ?” - Erroneous: Email: “falseemail” (Should contain a “@ symbol”, Should contain a domain e.g. “.com”), Password: “pass” (Should be longer, should be a string, should have a capital letter and a symbol) - Extreme: “test@testem ail.com”, Password: QwE12? (Exactly 6 characters which is the minimum)
03/02/2026	log out function	Black box testing, Functionality testing, Security testing, Session management testing.	No test data required, the log in system must be fully functional and allow users to create accounts and save their data to a database and the log out button should be clear and easily visible.
19/02/2026	Log in button on the top right of the home page on the navigation bar	Black box testing, Functionality Testing, Intergration Testing	No test data required. The register page and form should have been created and fully functional. The tester should click the button to ensure it loads the correct page.
23/02/2026	Book a room	Black box testing, White box testing, Functionality testing, Security testing, Session management testing.	Test data required. The models and controller logic must have been created and there must be a logged in user to test it. The tester should book a room and fill

			out all fields and make sure it successfully logs the booking and displays it in the correct place. They should be able to edit, delete and view the details of the booking. Normal: Name: “John Doe”, Check-in: “2026-02-15 14:00”, Room: “Room 1”, Special Requests: “High floor, near elevator”, Number of Guests: “2” - Erroneous: Name: “12345” (Should contain letters), Check-in: “15-02-26” / “yesterday” (Wrong format / Past date not allowed), Room: “Room 999” (Non-existent), Special Requests: >500 characters (Should be ≤500), Number of Guests: “0” / “-1” / “abc” (Minimum 1, must be numeric) - Extreme: Name: “A…A” (100 characters), Check-in: “2100-12-31 23:59” (Far future), Room: “999999999” (Very large ID), Special Requests: Exactly 500 characters (Maximum), Number of Guests: “50” (Maximum allowed)

27/02/2026	Pay for booking	Black box testing, White box testing, Functionality testing, Security testing, Session management testing.	Test data required. The models and controller logic must have been created and there must be a logged in user to test it. The tester should book a room and fill out all fields and try to pay for a booking to test if the payment functions work properly. The payment should go through and finances should be transferred successfully. Normal: Name: “John Doe”, Credit Card Number: “4111 1111 1111 1111”, CVV: “123” - Erroneous: Name: “12345” (Should contain letters), Credit Card Number: “1234 5678 9012 345” / “abcd efgh ijkl mnop” (Too short / Non-numeric), CVV: “12” / “abcd” / “1234” (Should be exactly 3 numeric digits) - Extreme: Name: 100-character string, Credit Card Number: “9999 9999 9999 9999” (Max digits), CVV: “000” (Edge case, technically valid)
